MIL-P-60603(MU)
10 August 1966
SUPERSEDING
FA-PD-MI-2489, Rev. 1
15 November 1963

MILITARY SPECIFICATION

PROJECTILE, AP-T, 76MM, M339
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification is the applicable detail specification for Projectile, AP-T, 76MM, M339, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-10025	- Packaging, Packing, and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for
MIL-M-11472	- Magnetic Particle Inspection, Process for Ferromagnetic Materials

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Method

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Certification of Magnetic Particle Inspection Personnel
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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2.1 (Cont'd)

DRAWINGS

Munitions Command
F-7548604 - Pallet, for Fixed and Semi-Fixed Projectile, Packaging and Marking
D-10520209 - Metal Parts Assembly
IEL-10520209 - Index of Inspection Equipment List for Projectile, AP-T, 76MM, M339 Metal Parts Assembly

PUBLICATIONS

Munitions Command
Operating Instruction for Use of Detroit Testing for Band Seating Inspection
PA-PD-792 - Equipment, Testing or Measuring, for Ammunition Items

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirement specified on the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.3 Projectile body.-

3.3.1 Metal defects.- The projectile body shall contain no cracks in any portion of the body, imperfections or nonmetallic inclusions greater than 1/8 inches in length in the ogive contour, or imperfections or nonmetallic inclusions 2 inches in length or greater in any region of the projectile body.

3.4 Mechanical properties.

3.4.1 Body.- Mechanical properties and hardness shall be compatible with required terminal ballistics capability. Hardness pattern shall be established and maintained throughout production which will assure consistent defeat of required armor thickness (See 6.3).

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3.5 Body.

3.5.1 Chemical composition.- The composition of the steel used in shot bodies shall be at the option of the contractor, provided the shot complies with the function requirements and tests specified herein (See 6.3.1).

3.5.2 Engineering data.- An engineering data sheet shall be furnished with each lot and forwarded to the Proving Ground on a separate card. This card shall contain the following information:

a. Ladle analysis of each heat of steel

b. Heat treatment

c. Hardness pattern through the longitudinal section

3.6 Rotating band.- After seating, and subsequent removal from the projectile body, the rotating band shall have the following minimum mechanical properties: Tensile strength, 32,000 pounds per square inch (PSI) minimum; elongation, 15 percent minimum in one inch gage length.

3.6.1 Sidewall contact.- The seated band blank shall make contact with the band seat sidewall for a minimum of $\frac{1}{2}$ of the band seat depth measured in a direction from the body outside diameter toward the bottom of the band seat.

3.7 Rotating band, blank.- The rotating band blank shall conform to the requirements of the applicable drawing. The blank shall be annealed if necessary, to comply with the requirements of the applicable specification, except that in case of hot banding at not less than 800 degrees Fahrenheit (^{o}F) no prior annealing will be necessary. The band shall be free of foreign material.

3.8 Application of rotating band.- The band may be applied either hot or cold, but if applied hot, the contractor shall perform the banding operation in a manner which will hold the scale between the band and the band seat to a minimum. The bands shall not be heated above 1600° F to prepare them for application to the projectile. The bands shall be heated to the same degree throughout and in a manner which will prevent undue oxidation (loose scale).

3.8.1 Band seating.- The diametrical clearance between rotating band and band seat shall not exceed .006 inch.

3.9 Protective coating.- Protective coating shall be in accordance with the requirements of the applicable drawings and specifications referenced thereon.

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3.10 Proving Ground functional tests.- The assembled projectile shall reveal no evidence of failure when tested at excess and service pressure. Metal parts failure shall be:

a. Separation or breakup of components.

3.10.1 Penetration.- The protective ballistic limit (PBL) of the sample shall not exceed the PBL of the reference lot by more than 150 feet per second (FPS).

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.11.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.11.3 Foreign matter.- No part or assembly shall contain dirt, grease, chips, rust, or other foreign matter. Discoloration not due to corrosive action will be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspections required by the drawings and specification shall be considered as a condition precedent to performing the Proving Grounds functional tests (see 3.10).

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b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Grounds functional tests in 4.3.4.

4.2 First Article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 41 metal parts as follows:

31 metal parts assemblies complete (painted)
3 metal parts assemblies complete (unpainted) except that the rotating band shall have a light surface cut to remove banding indentations
2 metal parts assemblies complete (unpainted)
5 sets of component parts

Twenty-nine (29) of the above painted metal parts shall be forwarded to a designated proving ground for ballistic tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used for regular production. All parts and materials including packaging and packing shall be the same as used for regular production, and shall be from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The supplier shall submit the product in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235 as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Windshield, Drawing D10520212 covering a detail of Drawing D10520209.

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter outside large	Gage
102. Diameter inside large	Gage
103. Length to forward (small) inside datum diameter	Gage
104. Length to center inside basic diameter	Gage
105. Length to rear inside basic diameter	Gage
106. Length of ogive seat	Gage
107. Length to forward outside basic diameter	Gage
108. Length to rear outside basic diameter	Gage
109. Metal defective	Visual
MINOR	
201. Length total	Gage
202. Surface finish, improper	Visual (see 6.2)
203. Burr	Visual
204. Foreign matter	Visual

4.3.2.2 Body covering a detail of Drawing D10520209.

CRITICAL	
None defined	
MAJOR	
101. Diameter of rotating band seat	Gage
102. Width of rotating band seat	Gage
103. Scoring missing from rotating band seat	Visual
104. Foreign matter on rotating band seat	Visual
MINOR	
201. Location of rotating band seat	Gage
202. Fillet(s) missing from rotating band seat	Visual

4.3.2.3 Body Assembly, Drawing F10520210 covering a detail of Drawing D10520209.

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of nose with bourrelets	Gage
102. Length from rear face to forward basic diameter	Gage
103. Length from forward basic diameter to rear basic diameter	Gage
MINOR	
201. Width of nose flat, max	Gage

4.3.2.4 Metal parts assembly, Drawing D10520209 (prior to painting).

CRITICAL	
None defined	
MAJOR	
101. Concentricity of windshield with bourrelets	Gage
102. Concentricity of body with bourrelets	Gage
103. Concentricity of rotating band lip with bourrelets	Gage
104. Concentricity of rotating band flats with bourrelets	Gage
105. Concentricity at rear of body with bourrelets	Gage
106. Concentricity of tracer cavity with bourrelets	Gage
107. Tracer cavity not functionally concentric with minor diameter of thread	Gage
108. Diameter of bourrelets, min.	Gage
109. Diameter of rotating band lip	Gage
110. Diameter of rotating band flats	Gage
111. Diameter of rotating band grooves	Gage
112. Length from rear of rotating band lip to start of angle	Gage
113. Width of forward rotating band groove	Gage
114. Width of rotating band	Gage
115. Width of rear rotating band groove	Gage

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4.3.2.4 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of body (between bourrelets)	Gage
202.	Diameter at rear of rotating band	Gage
203.	Diameter of chamfer at tracer cavity	Gage
204.	Location of forward bourrelet	Gage
205.	Location of rear bourrelet	Gage
206.	Location of rotating band	Gage
207.	Location of rotating band groove	Gage
208.	Depth of tracer cavity drill point, max (if applicable)	Gage
209.	Depth from rear face to start of tracer cavity	Gage
210.	Depth of undercut front and rear of rotating band (if applicable)	Visual (see 6.2)
211.	Width of undercut front and rear of rotating band (if applicable)	Visual (see 6.2)
212.	Radii or chamfers missing or incorrect	Visual (see 6.2)
213.	Surface finish improper	Visual (see 6.2)
214.	Marking missing, incorrect or illegible	Visual
215.	Burr	Visual
216.	Foreign matter	Visual

4.3.2.5 Metal parts assembly, Drawing D10520209 (after painting).

CRITICAL		
1.	Magnetic particle stamp missing (indicating failure to comply with requirements)	Visual
MAJOR		
101.	Weight	Approved scale
102.	Pitch diameter of tracer thread, max	Gage
103.	Minor diameter of tracer thread, max	Gage
104.	Diameter of bourrelets, max	Gage
105.	Length of effective tracer thread, min	Gage
106.	Length total, max	Gage
107.	Insecure assembly	Visual and manual
108.	Tracer thread damaged or not full	Visual
109.	Rotating band damaged	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of tracer thread, min.	Gage
202. Diameter of tracer cavity	Gage
203. Depth of tracer cavity, max.	Gage
204. Presence of tracer thread undercut (if applicable)	Visual
205. Protective coating inadequate or damaged	Visual
206. Paint on rotating band bearing surfaces	Visual
207. Paint on threads	Visual
208. Dessicant missing from tracer cavity	Visual
209. Tracer cavity not properly sealed	Visual
210. Burr	Visual
211. Foreign matter	Visual

4.3.2.6 Plug and disc assembly, Drawing B8597489 covering a detail of Drawing B10520209.

CRITICAL	
None defined	
MAJOR	
101. Inside diameter not functionally concentric with pitch diameter of thread	Gage
102. Pitch diameter of thread, min.	Gage
103. Major diameter of thread, min.	Gage
104. Diameter of hole	Gage
105. Thickness of body	Gage
106. Thickness of disc	Gage
107. Threads damaged or not full	Visual
108. Improper assembly	Visual
109. Metal defective	Visual
110. Protective coating, inadequate or damaged	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Presence and location of wrench holes	Visual (see 6.2)
203. Surface finish, improper	Visual (see 6.2)
204. Burr	Visual
205. Foreign matter	Visual

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4.3.2.7 Carton (prior to sealing) (plug and disc assembly).

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Improper assembly	Visual
202. Support missing or damaged	Visual
203. Separator missing or damaged	Visual
204. Incorrect number in carton	Visual

4.3.2.8 Carton (after sealing) (plug and disc assembly).

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not properly sealed	Visual
203. Contents loose	Manual
204. Marking missing, incorrect or illegible	Visual

4.3.2.9 Pallet Drawing F7548604 (projectile).

CRITICAL

None defined

MAJOR

None defined

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve(s) missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual and manual
205. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Metal defects (major defects).- Each projectile body shall be subjected to a magnetic particle test. Failure of any projectile body to comply with 3.3.1 shall be cause for rejection of the body.

4.3.3.2 Mechanical properties.

4.3.3.2.1 Body.- One projectile body shall be selected from each heat of steel in the lot, to be used to develop the hardness pattern required 3.5.2.c.

4.3.3.2.2 Rotating band.- One projectile body assembly, after banding but prior to machining of the rotating band, shall be selected from each 8 hours production or fraction thereof for each banding machine in operation to determine compliance with 3.6. Failure of the band to comply shall be cause for rejection of the production represented by the sample.

4.3.3.3 Band seating.- Ten projectiles from each one hours production or fraction thereof per banding machine in operation shall be subjected to this test to determine compliance with 3.8.1.Failure to comply shall be cause for rejection of the lot.

4.3.3.4 Sidewall contact.- Rotating bands removed in accordance with 4.3.3.2.2 shall be used to determine extent of contact with sidewall of the band seat. Failure to comply with 3.6.1 shall be cause for rejection of the lot.

4.3.3.5 Protective coating.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.4 Proving Ground functional tests.- The following functional tests will be performed by the Government:

4.3.4.1 First article sample.

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4.3.4.1.1 Initial production sample.- The quantities listed in Table I shall be ballistically tested in the phase indicated.

TABLE I

Phase	Quantity
Penetration	9
Excess pressure	10
Service pressur	10

4.3.4.2 Production lot sample.- The quantities listed in Table II shall be ballistically tested in the phase indicated.

TABLE II

Phase	Quantity
Penetration	9
Excess pressure	5
Service pressure	2

4.3.4.3 Penetration.- The samples selected in accordance with Table I and Table II will be subjected to this test. Failure of the sample projectiles to comply with 3.10.1 will be cause for rejection of the lot.

4.3.4.4 Excess pressure.- Samples provided in accordance with Table I and Table II will be subjected to this test. Failure of any projectile assembly to comply with 3.10 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.5 Service pressure.- Samples provided in accordance with Table I and Table II will be subjected to this test. Failure of any projectile assembly to comply with 3.10 shall be cause for rejection of the initial production sample of the production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.- Index of Inspection Equipment Lists, IEL-10520209 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable IEL.

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4.4 Test methods and procedures.

4.4.1 Metal defects.- The projectile body shall be tested as prescribed in MIL-M-11472, except that the dry method and continuous method procedure shall not be used. A magnetizing current of 2500 amperes is required. Either direct current (DC) or alternating current (AC) may be used in magnetizing the sample under examination. When using the AC method, the current should be cut off at the first quarter (90°) of the alternating current cycle sine wave to insure that the body is magnetized. After acceptance, the letter "M" shall be stamped on the base to signify compliance with 3.3.1.

4.4.2 Mechanical properties.

4.4.2.1 Body.- The sample provided by 4.3.3.2.1 shall be sectioned longitudinally through the center. The hardness pattern required to be recorded on data card by 3.5.2.c shall be developed using a Rockwell Hardness Tester and the procedures for performing this test as prescribed by Federal Test Method Standard No. 151.

4.4.2.2 Rotating band.- Following the examination for depth of sidewall contact as described in 4.4.4, the rotating band shall be removed from one of two segments from the sample provided by 4.3.3.2.2 and carefully flattened without hammering so as to hold additional cold working to a minimum. A test specimen conforming to an "F" type as specified in Federal test method Standard No. 151 shall be removed and tested in accordance with procedures prescribed by Federal Test Method Standard No. 151.

4.4.3 Band seating.- The diametrical clearance between the rotating band and the band seat shall be tested using a Detroit Testing Machine (see 6.1).

4.4.4 Sidewall contact.- The samples provided by 4.3.3.4 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two parts shall be used for establishment of mechanical properties of the band; the other part shall be sectioned into two parts, thus exposing four surfaces, each of which reveals a veiw of the band as swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact. Visual examination shall be made of the exposed swaged surfaces to determine compliance with 3.6.1

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4.4.5 Protective coating.- Procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified on the applicable component drawing and in the specification referenced thereon.

4.4.6 Proving Ground functional test.

4.4.6.1 Penetration.- Prior to firing penetration samples, the protective ballistic limit (PBL) of a reserve reference lot of the same caliber and model will be determined. The PBL is defined as the mean of six velocities - to include three complete and three partial penetrations, the velocity spread of which shall not exceed 150 FPS. Following determination of the PBL of the reference lot, the PBL of the test lot shall be determined as indicated above utilizing samples selected in accordance with Table I. Sample projectile will be assembled into complete rounds utilizing a charge as determined by the reference lot firing. The rounds will be fired in a gun for which it is standard, against 2 inch thick homogeneous armor plate placed approximately 300 feet from the muzzle of the gun and at an obliquity of 60 degrees. A complete penetration will be considered to have occurred if a fragment of the plate or projectile is thrown beyond the rear face of the plate with sufficient energy to penetrate an 0.020 inch aluminum alloy (24ST) sheet, or its equivalent, placed parallel to and one foot behind the plate. When, in the Proof Officer's opinion, it is possible to observe that these conditions are being complied with without the use of a sheet, the sheet may be omitted. An impact which occurs under the following conditions will be disregarded and another sample fired in its place:

a. Any projectile which strkes within two calibers of the edge of the plate.

b. Any projectile which strikes within one caliber of a previous impact or whose path of penetration intersects a previous hole.

c. Any projectile which strikes the spalled area of a previous impact.

4.4.6.2 Excess pressure.- The projectiles supplied in accordance with 4.3.4 for this phase will be assembled into complete rounds and fired in a gun for which the round is standard. A propellant charge weight adjusted to produce 115 percent of the service pressure of the round in a new gun shall be used. The projectile will be fired through a canvas or heavy muslin screen placed approximately 1,000 meters from the muzzle of the gun. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile, for the initial production sample and the production lots, (prior to firing, paint will be applied to the windshield) (observation for paint on the cloth target will be made for windshield security).

4.4.6.3 Service pressure.- The projectiles supplied in accordance with 4.3.4 for this phase will be assembled into complete rounds utilizing the service charge and fired in a gun for which the round is standard for comparison with a like group from a reference projectile lot will be fired alternately through a canvas or heavy muslin screen placed approximately 1,000 meters

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4.4.6.3 (Cont'd)

from the muzzle of the gun. Velocity and accuracy measurements will be obtained for this range accuracy in terms of maximum spread in inches, both horizontal and vertical and probable error (PE) in mils horizontal and vertical will be recorded for all rounds fired and forwarded to the cognizant technical agency. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.

5.1.1 Level A.- Not applicable

5.1.2 Level B.- Not applicable

5.1.3 Level C.- Packing and marking shall be in accordance with Drawing F7548604.

5.1.4 Level C.- Plug and disc assembly, Drawing B8597489 shall be packed in accordance with MIL-P-10025. Pack 2,000 plug and disc assemblies in 2 layers arranged in a 10 x 10 pattern, 10 to a cell, using interlocking separators.

5.2 Data card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Band seating.- The operating procedures for use of the Detroit Testing Machine, as prepared by the Inspection Engineering Office, Frankford Arsenal, SMUFA-Q3000, may be used as an aid in checking band seating inspection procedures.

6.2 Visual examination qualification.- When a compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Heat treatment.- Hardness pattern of figure 1 is advisory and is representative of shot bodies which have satisfied the plate penetration requirements of 3.10.1.

6.3.1 4160 steel with the following chemistry and heat treatment has been used in post production to obtain advisory hardness pattern shown on figure 1:

Chemistry

C = 0.61	s = 0.035	cr = 1.00
MN = 0.91	s1 = 0.28	Mo = 0.21
A1 - 0.017	Ni = 0.00	Vo = 0.00

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6.3.1 (Cont'd)

Heat treatment

1650° F in salt, total time 49 min.
Quench in salt, 375° F 9 min.
Air cool to room temperature
Temperature 350° F, in oil - 3 hours
Base draw 1250° F - 9 min. in salt base immersed 3 inches deep.
Air cool to room temperature; nose immersed in water to bourrelet.

6.4 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1315-A539

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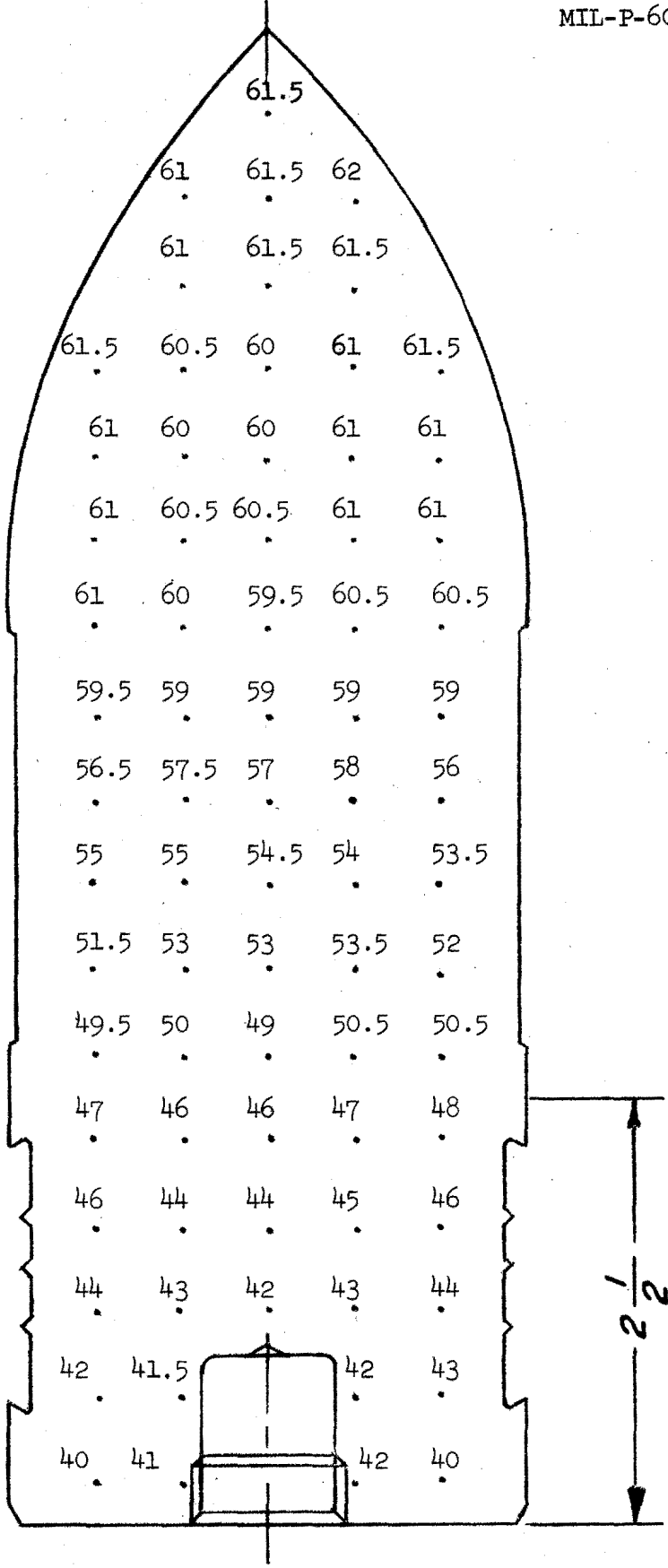


FIGURE I
Heat Treatment Hardness Pattern
for Projectile, AP-T, 76MM, M339
Metal Parts Assembly

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